

WORK EXPERIENCE

Shell International Exploration and Production Inc.

Intern in A.I. Natural Science Team

Houston, TX
May – August 2025

- Researched deep learning-based Full Waveform Inversion and PDE-solving neural operators (FNO, DeepONet, PINNs) leveraging **NVIDIA PhysicsNemo**.
- Built Devito-based data generation pipelines for high-fidelity forward and adjoint wavefield simulations.
- Enhanced baseline architectures and tuning strategies, **cutting MAE from 12.0 to 1.75**.
- Integrated advanced neural operators into the FWI pipeline, implementing steepest descent, conjugate gradient, and L-BFGS optimization schemes to build a **deployable hybrid pipeline**.

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- Worked on benchmarking State of the Art algorithms and techniques used for Semantic Segmentation under Limited Labels
- Implemented Parameter Efficient Training of Large Vision Transformer models like DinoV2 and SAM
- Created an **end-to-end deployable pipeline** to conduct 3D segmentation inference on seismic volume using our models

EDUCATION

Georgia Institute of Technology

Ph.D. in Electrical and Computer Engineering, Advisor: Dr. Ghassan AlRegib

Atlanta, GA
Fall 2022 - present

RESEARCH EXPERIENCE

Graduate Research Assistant, CeGP (Advisor: Dr. Ghassan AlRegib)

(Summer 2022 – present)

Towards Trustworthy AI via Causality, Information & Human Interaction:

- Proposed a novel causal approach to calculate necessity and sufficiency score to quantify the robustness and trust in local explanatory modules used for generating explanations in decisional making tasks.
- Proposed a new adversarial explanation approach to test robustness of objective causal metrics for XAI.
- Developed an information theoretic based modelling strategy to objectively quantify the trustworthiness of attribution-based decision making in vision models like classifiers, detectors and VLMs.
- Developed a novel optimization and active querying-based strategy to tackle human interaction (prompting) in large vision and language models to improve performances while reducing interaction costs.

Master's Thesis, AVA Lab (Advisor: Dr. Larry Heck)

(Spring & Fall 2022)

Visual and Tabular Question Answering in Conversational AI:

- Surveyed and implemented a retrieval based, LLM assisted multi-modal semantic question answering system trained on a corpus of Wikipedia texts, images, and tables.

PUBLICATIONS

- **P.Chowdhury**, et al. "Mind the Primitive Gap: Explanations for Interactive AI Should Guide Actions, Not Justify Outputs" in *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*. [[under blind submission](#)]
- **P. Chowdhury**, et al. "Tabular Data in Interactive and Conversational AI: A Survey of Foundations, Benchmarks, Systems and Open Problems" at *Transactions on Machine Learning Research (TMLR)*, 2026. [[under blind submission](#)]
- **P. Chowdhury** M. Prabhushankar, and G. AlRegib. "BALD-SAM: Disagreement-based Active Prompting in Interactive Segmentation." arXiv preprint arXiv:2603.10828 (2026).
- **P.Chowdhury** et. al. "An Information Theoretic Framework for Quantifying Prompt Uncertainty" at *Transactions on Machine Learning Research (TMLR)*, 2026. [[under blind submission](#)]
- **P.Chowdhury** et. al. "The Cost of Intervention. A Lipschitz Budget for Perturbation in XAI" at *European Conference for Computer Vision (ECCV)*, 2026. [[under blind submission](#)]
- **P. Chowdhury**, et al. "A unified framework for evaluating robustness of Machine Learning Interpretability for Prospect Risking." *Geophysics* 90.3 (2025)
- **P. Chowdhury**, et al. "Are Objective Explanatory Evaluation Metrics Trustworthy? An Adversarial Analysis," *IEEE International Conference on Image Processing (ICIP)*, Oct. 27 - 30, 2024.
- **P. Chowdhury**, et al. "Explaining Explainers: Necessity and Sufficiency in Tabular Data," in *NeurIPS 2023 Workshop: Table Representation Learning*, Nov. 15, 2023